

## **R Practice Project 2 – Exploratory Data Analysis (EDA) of 1986 MLB MVP Award**

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ALY 6000 Introduction to Analytics

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April 19, 2026

The purpose of ALY6000 Module 2 Project was to analyze statistics from the 1986 MLB season and, based on these data points, select a player that is deserving of the MVP award for that season in professional baseball. Important statistics that come into play when analyzing the performance of an MLB player include their on-base percentage (OBP), home runs (HR), runs batted in (RBI) – this coincides with how many games and at-bats these players actually take to ensure they have qualified for the awards threshold. Based on the assignment and the data analyzed in R Studio, I think that Boston Red Sox shortstop/third baseman Wade Boggs is deserving of the MLB Most Valuable Player (MVP) award for the year 1986.

There were many other players including Red Sox pitcher Roger Clemens (who won it originally) and Yankees first basemen Don Mattingly who also could've won MVP based on their performance – however, the process conducted in this project helped to filter out players and ultimately determine that Boggs had the best baseball season for the year 1986. After downloading the data of all eligible players for the 1986 season, any player with 0 at bats was filtered out immediately, because they didn't play and therefore would not qualify for MVP. The categories on-base percentage (OBP) and batting average (BA) were then added to our data for further analysis because of how important both these categories are when considering what a player accomplished in a given year. In order to get the most accurate results, there were other methods of filtering out players from that season that were implemented into our dataset. For example, the top 10 players that struck out the most were added to a separate variable called strikeout artist – this is because a player may bat many times and have lots of hits/good stats, but it is also a very big deal in terms of negative impact if they strike out during a high percentage of at bats. It would be hard to consider someone for MVP despite appealing

offensive numbers if they were listed in the top 10 in strikeouts throughout the entire MLB. Additionally, the data was filtered down to right around 250 players through applying the eligibility standards for winning an MVP in Major League Baseball. An individual player must participate in at least 100 games and also bat at least 300 times throughout the season in order to be considered for any of the end of season awards. It was definitely interesting that such a small percentage of the participating players were actually eligible for the MLB MVP award – there was right around 725 players that were listed on a team's roster/participated in a game for the 1986 baseball season, but only around 250 of those players met the criteria for qualifying to win the MVP award.

The key takeaway from this assignment is that Wade Boggs deserved to win the MVP award in 1986 because of how much better his stats were than any other participating, eligible players. Boggs had a batting average of 0.356 which is unheard of in the contemporary era of MLB performances. Furthermore, the only person within 0.20 of his batting average was Don Mattingly, who was batting 0.352, and Boggs has him beat in many other statistical categories for that season. The drop-off in average after Boggs and Mattingly went all the way down to 0.339 with Hubie Brooks. Wade Boggs was sixth in the league in hits with 207 hits, which was not overly impressive compared to other players such as Don Mattingly and Kirby Puckett who had a double-digit lead on Boggs with 238 hits and 223 hits respectively. However, the most impressive part of Boggs performance was that he was able to do all of this in under 600 plate appearances, where he sat at 580 at-bats for the entire season - other players near him in hits for the season had all batted at least 600 times. Boggs also deserves the MVP award for the 1986 MLB season because of certain stats that may go unnoticed but have a massive impact on

how a player and team performs throughout a given year. Out of all eligible players, Boggs led the league in walks (BB) with a triple-digit showing of 105 – this is mostly impressive because of how big of an impact it has on Boggs on-base percentage (OBP). It also shows that Boggs is such a good batter that the pitchers deferred to walking him over pitching to him directly because of his offensive potential during any at-bat. To elaborate on the stats that may go unnoticed but have a large impact, Boggs only struck out 44 times out of his 580 at bats for an incredible 7.6% strikeout rate while putting up MVP offensive statistics. This would mean that out of all 250 eligible players for the 1986 MLB season, Boggs was top 50 in not striking out to opposing pitchers. Being able to walk a lot and avoiding strikeouts is a huge part of baseball because it shows a good sign of hand-eye coordination and the ability to make split-second decisions. Boggs was able to get on base so often and have such a large impact on his team because he was rarely ever striking out and if he wasn't getting a hit, he was finding a way to get on base otherwise through walks (BB).

The counterargument against Wade Boggs 1986 MVP case is the underwhelming numbers that were put up in the Home Run (HR) and Runs Batted In (RBI) categories. Boggs 8 home runs (HR) put him at the bottom third tier for all eligible players – his 71 runs batted in is not unimpressive but rather leaves him in the middle tier of players with a mediocre performance regarding driving teammates in that are on base already. These statistics may have been the driving factor as to why Roger Clemens ultimately won the MVP award for the American League. However, Boggs advantage in the on-base percentage category is much too hard to overlook when comparing his performance to others from that 1986 season and considering him in the MVP race. Boggs on-base percentage of 0.455 was sitting over 0.40

higher than the next closer participant who was Tim Lincecum and had a 0.413 OBP – he also sat over 0.55 higher than Mattingly, who had an OBP of 0.398, thus eliminating the argument for Mattingly over Boggs. To interpret how OBP works, it essentially measures the frequency of how often a batter can reach base and essentially not get out. Boggs 0.455 OBP meant that in almost every other at bat ( $1/2$ ), he was able to reach base and provide offensive value to his team. Boggs was able to get on base quite often and when analyzing the histogram of the batting average data, it appears he is in the very small top quartile of players who batted right around 0.350 for the season.

Based on this data, it is pretty clear that Wade Boggs should have won the 1986 MLB MVP award – his OBP and BA are at the top of the league, and though his HR and RBI statistics were not very impressive, it was more the impact that he made on his team. Outside of the dataset, it is shown that Boggs WAR is 8.1 and would be the highest out of any respective player for that MLB season. Though WAR (Wins Above Replacement) didn't officially appear as a statistic until the early 2010's, it should be noted that this statistic is a comprehensive number designed to estimate all facets of the game that a player contributes to their team. Boggs WAR was listed as 8.1 for the 1986 MLB season and this is the highest in the league with Jesse Barfield (7.6) and Don Mattingly (7.2) trailing behind. Boggs impact to his team was so crucial that his 8.1 WAR statistic means he was that much better than an average replacement level player that would've been at his position during the 1986 MLB season.

To ultimately summarize and interpret the results of this project, it was mainly geared to analyze different baseball related variables for participating players during the 1986 MLB season. After determining the eligibility of each player, the most important categories such as

batting average (BA), on-base percentage (OBP), home runs (HR), and runs batted in (RBI) were analyzed to determine which player had the biggest impact on their team for that given season. Upon looking at the statistical datapoints, it was determined that because of his large advantage in BA and OBP, Wade Boggs had the biggest impact for his team in that 1986 MLB season and therefore, should have won the MVP award that year. In conclusion, this ALY6000 project was a great way of laying out the different statistical performances in R Studio, of each player in the MLB for 1986, and it was determined that Boggs had performed the best out of everyone involved.

## References

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Manning Publication. ISBN 978-1-617-29605-5